Lab Sheet 1 - Solution

Database Management System Lab (BTCS203P) | DDL Commands in MySQL

B.Tech Data Science, 3rd Semester

Objective: To understand and practice Data Definition Language (DDL) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

Note: All queries below were actually executed (on a MySQL-compatible server, MariaDB 10.11) and the output shown is the real result. Each task is a separate query, as the instructions require. Where a task changes structure silently, a verification query is shown right after it. The *Staff* table is dropped (2m) at the very end of the run, because Exercise 4(b) and 4(c) need it.

Task 1: Create a Table

1a. Create database Company

SQL

```
DROP DATABASE IF EXISTS Company;
CREATE DATABASE Company;
```

Verification query

```
SHOW DATABASES LIKE 'Company';
```

Output

```
+--------------------+
| Database (Company) |
+--------------------+
| Company            |
+--------------------+
```

1b. Create table Employees

SQL

```
USE Company;
CREATE TABLE Employees (
    EmpID      INT AUTO_INCREMENT PRIMARY KEY,
    FirstName  VARCHAR(50) NOT NULL,
    LastName   VARCHAR(50),
    Department VARCHAR(50),
    Salary     DECIMAL(10,2),
    Email      VARCHAR(100) UNIQUE
);
```

Verification query

```
SHOW TABLES;
```

Output

```
+-------------------+
| Tables_in_Company |
+-------------------+
| Employees         |
+-------------------+
```

Task 2: DDL Queries on the Employees Table

2a. Describe the structure of the table

SQL

```
DESCRIBE Employees;
```

Output

```
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| EmpID      | int(11)       | NO   | PRI | NULL    | auto_increment |
| FirstName  | varchar(50)   | NO   |     | NULL    |                |
| LastName   | varchar(50)   | YES  |     | NULL    |                |
| Department | varchar(50)   | YES  |     | NULL    |                |
| Salary     | decimal(10,2) | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  | UNI | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
```

2b. Rename the table to Staff

SQL

```
RENAME TABLE Employees TO Staff;
```

Verification query

```
SHOW TABLES;
```

Output

```
+-------------------+
| Tables_in_Company |
+-------------------+
| Staff             |
+-------------------+
```

2c. Add column PhoneNumber VARCHAR(15)

SQL

```
ALTER TABLE Staff ADD COLUMN PhoneNumber VARCHAR(15);
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| EmpID       | int(11)       | NO   | PRI | NULL    | auto_increment |
| FirstName   | varchar(50)   | NO   |     | NULL    |                |
| LastName    | varchar(50)   | YES  |     | NULL    |                |
| Department  | varchar(50)   | YES  |     | NULL    |                |
| Salary      | decimal(10,2) | YES  |     | NULL    |                |
| Email       | varchar(100)  | YES  | UNI | NULL    |                |
| PhoneNumber | varchar(15)   | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

2d. Modify Salary to DECIMAL(12,2)

SQL

```
ALTER TABLE Staff MODIFY Salary DECIMAL(12,2);
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| EmpID       | int(11)       | NO   | PRI | NULL    | auto_increment |
| FirstName   | varchar(50)   | NO   |     | NULL    |                |
| LastName    | varchar(50)   | YES  |     | NULL    |                |
| Department  | varchar(50)   | YES  |     | NULL    |                |
| Salary      | decimal(12,2) | YES  |     | NULL    |                |
| Email       | varchar(100)  | YES  | UNI | NULL    |                |
| PhoneNumber | varchar(15)   | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

2e. Add DEFAULT 'General' to Department

SQL

```
ALTER TABLE Staff ALTER COLUMN Department SET DEFAULT 'General';
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| EmpID       | int(11)       | NO   | PRI | NULL    | auto_increment |
| FirstName   | varchar(50)   | NO   |     | NULL    |                |
| LastName    | varchar(50)   | YES  |     | NULL    |                |
| Department  | varchar(50)   | YES  |     | General |                |
| Salary      | decimal(12,2) | YES  |     | NULL    |                |
| Email       | varchar(100)  | YES  | UNI | NULL    |                |
| PhoneNumber | varchar(15)   | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

2f. Drop the PhoneNumber column

SQL

```
ALTER TABLE Staff DROP COLUMN PhoneNumber;
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| EmpID      | int(11)       | NO   | PRI | NULL    | auto_increment |
| FirstName  | varchar(50)   | NO   |     | NULL    |                |
| LastName   | varchar(50)   | YES  |     | NULL    |                |
| Department | varchar(50)   | YES  |     | General |                |
| Salary     | decimal(12,2) | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  | UNI | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
```

2g. Add CHECK constraint (Salary > 0)

SQL

```
ALTER TABLE Staff ADD CONSTRAINT chk_Salary CHECK (Salary > 0);
```

Verification query

```
SELECT CONSTRAINT_NAME, CHECK_CLAUSE FROM information_schema.CHECK_CONSTRAINTS
WHERE CONSTRAINT_SCHEMA = 'Company' AND TABLE_NAME = 'Staff';
```

Output

```
+-----------------+--------------+
| CONSTRAINT_NAME | CHECK_CLAUSE |
+-----------------+--------------+
| chk_Salary      | `Salary` > 0 |
+-----------------+--------------+
```

2h. Rename column FirstName to GivenName

SQL

```
ALTER TABLE Staff CHANGE COLUMN FirstName GivenName VARCHAR(50) NOT NULL;
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| EmpID      | int(11)       | NO   | PRI | NULL    | auto_increment |
| GivenName  | varchar(50)   | NO   |     | NULL    |                |
| LastName   | varchar(50)   | YES  |     | NULL    |                |
| Department | varchar(50)   | YES  |     | General |                |
| Salary     | decimal(12,2) | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  | UNI | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
```

2i. Drop the chk_Salary constraint

SQL

```
ALTER TABLE Staff DROP CONSTRAINT chk_Salary;
```

Verification query

```
SELECT COUNT(*) AS remaining_check_constraints FROM information_schema.CHECK_CONSTRAINTS
WHERE CONSTRAINT_SCHEMA = 'Company' AND TABLE_NAME = 'Staff';
```

Output

```
+-----------------------------+
| remaining_check_constraints |
+-----------------------------+
|                           0 |
+-----------------------------+
```

2j. Create an index on Email

SQL

```
CREATE INDEX idx_Email ON Staff(Email);
```

Verification query

```
SELECT INDEX_NAME, COLUMN_NAME, NON_UNIQUE FROM information_schema.STATISTICS
WHERE TABLE_SCHEMA='Company' AND TABLE_NAME='Staff';
```

Output

```
+------------+-------------+------------+
| INDEX_NAME | COLUMN_NAME | NON_UNIQUE |
+------------+-------------+------------+
| PRIMARY    | EmpID       |          0 |
| Email      | Email       |          0 |
| idx_Email  | Email       |          1 |
+------------+-------------+------------+
```

2k. Drop the index on Email

SQL

```
DROP INDEX idx_Email ON Staff;
```

Verification query

```
SELECT INDEX_NAME, COLUMN_NAME, NON_UNIQUE FROM information_schema.STATISTICS
WHERE TABLE_SCHEMA='Company' AND TABLE_NAME='Staff';
```

Output

```
+------------+-------------+------------+
| INDEX_NAME | COLUMN_NAME | NON_UNIQUE |
+------------+-------------+------------+
| PRIMARY    | EmpID       |          0 |
| Email      | Email       |          0 |
+------------+-------------+------------+
```

2l. Truncate the table (remove all rows, keep structure)

SQL

```
INSERT INTO Staff (GivenName, LastName, Salary, Email) VALUES
  ('Amit', 'Sharma', 50000, 'amit@co.com'),
  ('Riya', 'Verma',  60000, 'riya@co.com');
SELECT COUNT(*) AS rows_before FROM Staff;
TRUNCATE TABLE Staff;
```

Output

```
+-------------+
| rows_before |
+-------------+
|           2 |
+-------------+
```

Verification query

```
SELECT COUNT(*) AS rows_after FROM Staff;
DESCRIBE Staff;
```

Output

```
+------------+
| rows_after |
+------------+
|          0 |
+------------+
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| EmpID      | int(11)       | NO   | PRI | NULL    | auto_increment |
| GivenName  | varchar(50)   | NO   |     | NULL    |                |
| LastName   | varchar(50)   | YES  |     | NULL    |                |
| Department | varchar(50)   | YES  |     | General |                |
| Salary     | decimal(12,2) | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  | UNI | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
```

2m. Drop the Staff table completely

SQL

```
DROP TABLE Staff;
```

Verification query

```
SHOW TABLES;
```

Output

```
+-------------------+
| Tables_in_Company |
+-------------------+
| Projects          |
+-------------------+
```

3. Conceptual Questions

1. What is the difference between TRUNCATE and DELETE?

DELETE is a DML command that removes rows one by one, can use a `WHERE` clause to remove only some rows, fires row-level triggers, can be rolled back inside a transaction, and does *not* reset the AUTO_INCREMENT counter. **TRUNCATE** is a DDL command that removes *all* rows at once by dropping and re-creating the table's data, so it has no `WHERE` clause, is much faster, resets AUTO_INCREMENT to 1, and cannot be rolled back (it causes an implicit commit in MySQL). Both keep the table structure.

2. Explain the use of the CHECK constraint.

A CHECK constraint enforces a condition on the values of a column (or row) so that invalid data is rejected at INSERT/UPDATE time. For example, `CHECK (Salary > 0)` stops negative or zero salaries and `CHECK (Budget > 0)` does the same for project budgets. It keeps data valid at the database level instead of relying only on application code. (MySQL enforces CHECK from version 8.0.16; older versions parsed but ignored it.)

3. Why is it important to define a PRIMARY KEY for a table?

A PRIMARY KEY uniquely identifies every row: it cannot be NULL and cannot repeat. This guarantees entity integrity, avoids duplicate records, gives a fast index for lookups, and is the column that other tables reference through FOREIGN KEYs to build relationships. Without it, rows cannot be reliably identified, updated or deleted individually.

4. How does ALTER TABLE help in modifying an existing table?

ALTER TABLE changes the structure of a table without recreating it or losing its data. It can ADD, DROP, MODIFY or CHANGE (rename) columns, add or drop constraints (PRIMARY KEY, UNIQUE, CHECK, FOREIGN KEY), set or remove default values, add or drop indexes, and rename the table. Examples from this lab: `ADD COLUMN PhoneNumber`, `MODIFY Salary DECIMAL(12,2)`, `DROP CONSTRAINT chk_Salary`.

5. What is the purpose of creating an index in a table?

An index is a separate sorted data structure (usually a B-tree) on one or more columns that lets the database find rows without scanning the whole table. It speeds up SELECT queries with WHERE, JOIN, ORDER BY and GROUP BY on the indexed columns. The trade-off is extra storage and slightly slower INSERT/UPDATE/DELETE because the index must be maintained. A UNIQUE index (like the one on Email) also enforces uniqueness.

4. Exercises

Exercise 4.1: Projects table. (a) 7 DDL operations performed (minimum asked: 5); (b) columns of Staff; (c) Department allowed to be NULL.

4.1 Create table Projects

SQL

```
CREATE TABLE Projects (
    ProjectID   INT PRIMARY KEY AUTO_INCREMENT,
    ProjectName VARCHAR(100) NOT NULL,
    StartDate   DATE,
    EndDate     DATE,
    Budget      DECIMAL(15,2) CHECK (Budget > 0)
);
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| ProjectID   | int(11)       | NO   | PRI | NULL    | auto_increment |
| ProjectName | varchar(100)  | NO   |     | NULL    |                |
| StartDate   | date          | YES  |     | NULL    |                |
| EndDate     | date          | YES  |     | NULL    |                |
| Budget      | decimal(15,2) | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

4.1a-(1) Add column Status

SQL

```
ALTER TABLE Projects ADD COLUMN Status VARCHAR(20);
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| ProjectID   | int(11)       | NO   | PRI | NULL    | auto_increment |
| ProjectName | varchar(100)  | NO   |     | NULL    |                |
| StartDate   | date          | YES  |     | NULL    |                |
| EndDate     | date          | YES  |     | NULL    |                |
| Budget      | decimal(15,2) | YES  |     | NULL    |                |
| Status      | varchar(20)   | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

4.1a-(2) Modify ProjectName to VARCHAR(150)

SQL

```
ALTER TABLE Projects MODIFY ProjectName VARCHAR(150) NOT NULL;
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| ProjectID   | int(11)       | NO   | PRI | NULL    | auto_increment |
| ProjectName | varchar(150)  | NO   |     | NULL    |                |
| StartDate   | date          | YES  |     | NULL    |                |
| EndDate     | date          | YES  |     | NULL    |                |
| Budget      | decimal(15,2) | YES  |     | NULL    |                |
| Status      | varchar(20)   | YES  |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
```

4.1a-(3) Set DEFAULT 'Planned' on Status

SQL

```
ALTER TABLE Projects ALTER COLUMN Status SET DEFAULT 'Planned';
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| ProjectID   | int(11)       | NO   | PRI | NULL    | auto_increment |
| ProjectName | varchar(150)  | NO   |     | NULL    |                |
| StartDate   | date          | YES  |     | NULL    |                |
| EndDate     | date          | YES  |     | NULL    |                |
| Budget      | decimal(15,2) | YES  |     | NULL    |                |
| Status      | varchar(20)   | YES  |     | Planned |                |
+-------------+---------------+------+-----+---------+----------------+
```

4.1a-(4) Rename column Status to ProjectStatus

SQL

```
ALTER TABLE Projects CHANGE COLUMN Status ProjectStatus VARCHAR(20) DEFAULT 'Planned';
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| ProjectID     | int(11)       | NO   | PRI | NULL    | auto_increment |
| ProjectName   | varchar(150)  | NO   |     | NULL    |                |
| StartDate     | date          | YES  |     | NULL    |                |
| EndDate       | date          | YES  |     | NULL    |                |
| Budget        | decimal(15,2) | YES  |     | NULL    |                |
| ProjectStatus | varchar(20)   | YES  |     | Planned |                |
+---------------+---------------+------+-----+---------+----------------+
```

4.1a-(5) Create an index on ProjectName

SQL

```
CREATE INDEX idx_ProjectName ON Projects(ProjectName);
```

Verification query

```
SELECT INDEX_NAME, COLUMN_NAME, NON_UNIQUE FROM information_schema.STATISTICS
WHERE TABLE_SCHEMA='Company' AND TABLE_NAME='Projects';
```

Output

```
+-----------------+-------------+------------+
| INDEX_NAME      | COLUMN_NAME | NON_UNIQUE |
+-----------------+-------------+------------+
| PRIMARY         | ProjectID   |          0 |
| idx_ProjectName | ProjectName |          1 |
+-----------------+-------------+------------+
```

4.1a-(6) Drop column EndDate

SQL

```
ALTER TABLE Projects DROP COLUMN EndDate;
```

Verification query

```
DESCRIBE Projects;
```

Output

```
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| ProjectID    | int(11)      | NO   | PRI | NULL    | auto_increment |
| ProjectName  | varchar(150) | NO   | MUL | NULL    |                |
| StartDate    | date         | YES  |     | NULL    |                |
| Budget       | decimal(15,2)| YES  |     | NULL    |                |
| ProjectStatus| varchar(20)  | YES  |     | Planned |                |
+--------------+--------------+------+-----+---------+----------------+
```

4.1a-(7) Rename table Projects to ProjectList, then back

SQL

```
RENAME TABLE Projects TO ProjectList;
RENAME TABLE ProjectList TO Projects;
```

Verification query

```
SHOW TABLES;
```

Output

```
+------------------+
| Tables_in_Company |
+------------------+
| Projects         |
| Staff            |
+------------------+
```

4.1b Retrieve names of all columns in the Staff table

SQL

```
SELECT COLUMN_NAME
FROM information_schema.COLUMNS
WHERE TABLE_SCHEMA = 'Company' AND TABLE_NAME = 'Staff'
ORDER BY ORDINAL_POSITION;
```

Output

```
+-------------+
| COLUMN_NAME |
+-------------+
| EmpID       |
| GivenName   |
| LastName    |
| Department  |
| Salary      |
| Email       |
+-------------+
```

4.1c Modify Department to allow NULL values

SQL

```
ALTER TABLE Staff MODIFY Department VARCHAR(50) NULL;
```

Verification query

```
DESCRIBE Staff;
```

Output

```
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| EmpID      | int(11)       | NO   | PRI | NULL    | auto_increment |
| GivenName  | varchar(50)   | NO   |     | NULL    |                |
| LastName   | varchar(50)   | YES  |     | NULL    |                |
| Department | varchar(50)   | YES  |     | NULL    |                |
| Salary     | decimal(12,2) | YES  |     | NULL    |                |
| Email      | varchar(100)  | YES  | UNI | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
```

Submission checklist

1. Screenshots of each step's output (run the attached .sql in MySQL Workbench / CLI and screenshot each result).
2. SQL script saved in a text file: **Lab_Sheet_1_Solution.sql**.
3. Submit both.